# Circularly polarised 144 MHz emission from the radio-flaring triple b Persei

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**Abstract**

b Persei (HD 26961, A1 III, 117 pc) has been known as a radio source since 1973, a triple whose inner pair is an ellipsoidal variable on a 1.53 d orbit and whose flares at gigahertz frequencies are non-thermal. The third data release of the LOFAR Two-metre Sky Survey holds a 144 MHz component 0.3″ from the star, and its per-pointing Stokes V images of four epochs from 2016 to 2022 give $+0.44\pm0.09$ mJy beam$^{-1}$ at the star's position ($5.1\sigma$), consistent between epochs; on 2018 November 21 Stokes I and V stand at $0.75\pm0.15$ and $+0.68\pm0.15$ mJy beam$^{-1}$, a circularly polarised fraction of $0.91 \pm 0.27$. Among the 204 magnetic chemically peculiar stars of the Renson and Manfroid catalogue within 300 pc that the images cover, b Per is the only one above $4\sigma$, and none of 816 positions 22.5″ away reaches it. The source's spectral luminosity, $1.2\times10^{16}$ erg s$^{-1}$ Hz$^{-1}$, and its brightness temperature, above $1.6 \times 10^{11}$ K for any region smaller than the inner orbit, require coherent emission. TESS gives the inner orbit's period as 1.5273435 d, and the four epochs cover the orbit from phase 0.45 through 0.12 with Stokes V positive in each. Gaia's velocity amplitude puts the inner companion above 0.35 $M_\odot$, and the ROSAT source on the system, $7.2\times10^{30}$ erg s$^{-1}$, is 20 to 140 times the saturated X-ray output of a main-sequence star of that mass. The emission reads as the electron-cyclotron maser of an active, evolved companion locked to the orbit, as in Algol and RS CVn binaries, with the A star's own magnetosphere as the alternative its classification as an Si star allows.

*Keywords:* Radio continuum: stars, Stars: individual: b Persei, Binaries: close, Stars: activity, Masers, Polarization

## 1. Introduction

Radio emission from close binaries with a cool, tidally locked active component has been studied at gigahertz frequencies since the 1970s, when b Persei was found as an intermittent 8.085 GHz source [1] and shown to be a triple with weak, non-periodic, non-thermal flares [2]. At 144 MHz, LOFAR hears active stars by their coherent emission, strongly circularly polarised: active M dwarfs [3] and RS CVn binaries [4], catalogued over a quarter of the northern sky by the circularly polarised survey of LOFAR's second data release (V-LoTSS) [5]. Two magnetic chemically peculiar stars were detected there in total intensity, with circular polarisation below 60 per cent [6].

The third data release of the LOFAR Two-metre Sky Survey (LoTSS-DR3) covers 88 per cent of the northern sky and publishes a Stokes V image of each pointing's epoch [7]. A measurement of those images at the magnetic chemically peculiar stars and the planet hosts of the northern sky found one star above $4\sigma$, b Per, which lies outside V-LoTSS's sky. This note reports b Per at 144 MHz over the four DR3 epochs that cover it, with its inner orbit from TESS and the other bands' archival data.

### 1.1. Contributions

1. The first detection of b Per at 144 MHz: a LoTSS-DR3 component on the star and circularly polarised emission in its per-pointing images, $5.1\sigma$ over four epochs and a fraction near one in the brightest.
2. The inner orbit's ephemeris from three TESS sectors of 2019–2024, the orbital phases of the four epochs, the companion's minimum mass from Gaia's velocity amplitude, and the system's X-ray luminosity set against what such a companion can emit.

## 2. b Persei

b Per (HD 26961, HR 1324, HIP 20070) is a naked-eye A star of $V = 4.6$, classified A1 III among spectroscopic binaries [8] and listed as an A2 Si star in the catalogue of Ap, HgMn and Am stars [9]; studies of the rotation of magnetic chemically peculiar stars include it [10, 11], and no measurement of its magnetic field is found in the literature. Its light varies as an ellipsoidal variable [12]. Hill et al. found the triple: the A star with an unseen companion on a 1.53 d orbit, and a third star on an orbit of about 700 d [2]. Gaia DR3 [13] gives a parallax of $8.547 \pm 0.233$ mas ($117 \pm 3$ pc), $G = 4.58$, $G_{\mathrm{BP}} - G_{\mathrm{RP}} = 0.13$, a renormalised unit weight error of 2.74, and a robust radial-velocity amplitude of $68.1\,\mathrm{km\,s^{-1}}$ over 41 transits; with $K_1 = 34.1\,\mathrm{km\,s^{-1}}$ on the 1.527 d orbit the mass function is $0.0063\,M_\odot$, and for an A star of $2.3\,M_\odot$ the companion's mass is at least $0.35\,M_\odot$. The ROSAT All-Sky Survey's bright-source catalogue [14] holds 1RXS J041814.8+501747, $4.2''$ from the star within its $7''$ error, at $0.469 \pm 0.031\,\mathrm{counts\,s^{-1}}$ and a hardness ratio HR1 of $0.19 \pm 0.06$, and the star is among the A stars of that survey whose X-rays Schröder and Schmitt studied [15]. It is one of the radio stars observed to link the Hipparcos frame to the radio frame [20].

## 3. Data and method

*LoTSS-DR3.* Each DR3 pointing is an 8 h observation of one epoch. The ASTRON VO's table of DR3 pointings gives the midpoint of each observation, and the SURF Data Repository holds each pointing's products as tar archives, from which the $6''$ dirty Stokes I and V planes were read by HTTP byte range, a patch around the star alone, and put on DR3's flux scale by the factor the release gives for the pointing. The search that found b Per measured Stokes V at the Si, Sr, Cr, Eu and He-peculiar stars of the Renson and Manfroid catalogue north of $-1^\circ$ within 300 pc and at the NASA Exoplanet Archive's hosts within 100 pc, each star carried to the pointing's epoch in its two nearest pointings, its noise from the robust scatter of the annulus $12''$–$30''$ out, positions $22.5''$ away serving as the null: 204 peculiar stars at a median $5\sigma$ limit of $0.53\,\mathrm{mJy\,beam^{-1}}$ and 353 hosts at $0.49\,\mathrm{mJy\,beam^{-1}}$. For b Per every pointing within $3^\circ$ was read, with positions $1'$ north and south as controls, and the source density of its field was counted in a $10'$ patch of its brightest epoch, local maxima above $4\sigma$ with the star excluded.

*TESS.* The two-minute PDCSAP light curves of sectors 19, 59 and 86 (2019–2024), 44685 cadences of quality zero, each normalised to its median, were fitted with the inner orbit's frequency and its first two harmonics and one level and slope per sector over a fine grid of periods, the period's error from the curvature of $\chi^2$ with the residual scatter as the noise. Sector 59 carries the slow drifts of a saturated star's photometry, and the shape of the curve is read from sector 19. The ephemeris for phases places phase zero at the deeper minimum.

Table 1: b Per in the per-pointing images of LoTSS-DR3: the pointing, the date of the observation's midpoint, the star's distance from the pointing centre, DR3's flux-scale factor, the phase range of the 8 h observation on the TESS ephemeris of the inner orbit, the dirty Stokes I peak within 4.5″ of the star and its offset, and Stokes V at that peak and at the star's position, all on DR3's flux scale.

| Pointing | Date | Offset (°) | Scale | Phase | $I$ (mJy beam$^{-1}$) | (″) | $V_{\rm peak}$ | $V_{\rm star}$ |
|---|---|---|---|---|---|---|---|---|
| P063+52 | 2016 Nov 10 | 1.42 | 0.957 | 0.75–0.97 | $0.12 \pm 0.15$ | 4.5 | +0.42 | $+0.31 \pm 0.14$ |
| P063+49 | 2018 Nov 21 | 1.44 | 1.027 | 0.90–0.12 | $0.75 \pm 0.15$ | 0.0 | +0.68 | $+0.68 \pm 0.15$ |
| P067+49 | 2018 Dec 1 | 2.24 | 1.021 | 0.45–0.67 | $1.04 \pm 0.29$ | 1.5 | +0.26 | $+0.24 \pm 0.21$ |
| P067+52 | 2022 May 7 | 2.31 | 1.053 | 0.53–0.75 | $0.80 \pm 0.33$ | 4.5 | −0.09 | $+0.45 \pm 0.30$ |

*Other bands.* The VLA Sky Survey's quick-look images of three epochs were read at the Canadian Astronomy Data Centre, the brightest pixel within 2.5″ of the star's position at each date against the image noise and against the same statistic at 2000 random positions 15″–150″ away. The ROSAT count rate is converted to flux in 0.1–2.4 keV with the hardness-dependent factor of Schmitt et al. [16].

## 4. Results

### *4.1. The 144 MHz source*

The DR3 catalogue holds ILTJ041814.68+501742.5, of peak $0.42 \pm 0.08$ mJy beam$^{-1}$ and integrated $0.79 \pm 0.22$ mJy, 0.3″ from the star's position at DR3's mean epoch. In the star's own field the images hold 394 sources per square degree at least as bright as the star's, so a source falls within 1.5″ of a given position by chance with probability $2.1 \times 10^{-4}$. In the search, b Per is the only one of the 204 peculiar stars above $4\sigma$ in Stokes V, and none of the 816 positions 22.5″ from them reaches $4\sigma$.

Table 1 gives the four epochs. On 2018 November 21 the dirty images hold a Stokes I peak of $0.75 \pm 0.15$ mJy beam$^{-1}$ on the star and Stokes V of $+0.68 \pm 0.15$ mJy beam$^{-1}$ there, a circularly polarised fraction of $0.91 \pm 0.27$. Ten days later Stokes I reaches $1.04 \pm 0.29$ mJy beam$^{-1}$ 1.5″ from the star, with Stokes V of $+0.24 \pm 0.21$ mJy beam$^{-1}$ at the star's position; in 2016 and 2022 Stokes I stands below $3\sigma$ and Stokes V at $+0.31 \pm 0.14$ and $+0.45 \pm 0.30$ mJy beam$^{-1}$. Read at the star's position, the four epochs give $+0.44 \pm 0.09$ mJy beam$^{-1}$ ($5.1\sigma$), consistent with one value ($\chi^2 = 4.6$ for 3 degrees of freedom), and the three without November 21 give $+0.31 \pm 0.11$ mJy beam$^{-1}$ ($2.8\sigma$). The eight control reads 1′ north and south of the star stay within $1.0\sigma$.

At 117 pc the November source has a spectral luminosity of $1.2 \times 10^{16}$ erg s$^{-1}$ Hz$^{-1}$, and the mean circularly polarised flux one of $7 \times 10^{15}$ erg s$^{-1}$ Hz$^{-1}$. Emission confined to a region the size of the inner orbit, $7.9\,R_\odot$ for the masses above, would have a brightness temperature of $1.6 \times 10^{11}$ K, and one the size of a $0.55\,R_\odot$ star $3 \times 10^{13}$ K.

### *4.2. Other bands*

In the VLA Sky Survey's quick-look images the star's position holds 0.25 mJy beam$^{-1}$ ($1.8\sigma$) on 2019 March 5, $0.71 \pm 0.21$ mJy beam$^{-1}$ ($3.3\sigma$) 0.5″ from it on 2021 November 13, a level random positions of the image reach in 1.4 per cent of cases, and 0.27 mJy beam$^{-1}$ ($2.1\sigma$) on 2024 May 25; NVSS at 1.4 GHz and TGSS at 150 MHz hold nothing within 30″, at limits of a few and about twenty millijanskys. The ROSAT rate gives $4.4 \times 10^{-12}$ erg cm$^{-2}$ s$^{-1}$, an X-ray luminosity of $7.2 \times 10^{30}$ erg s$^{-1}$. The Güdel–Benz relation of active stars [17] pairs that luminosity with

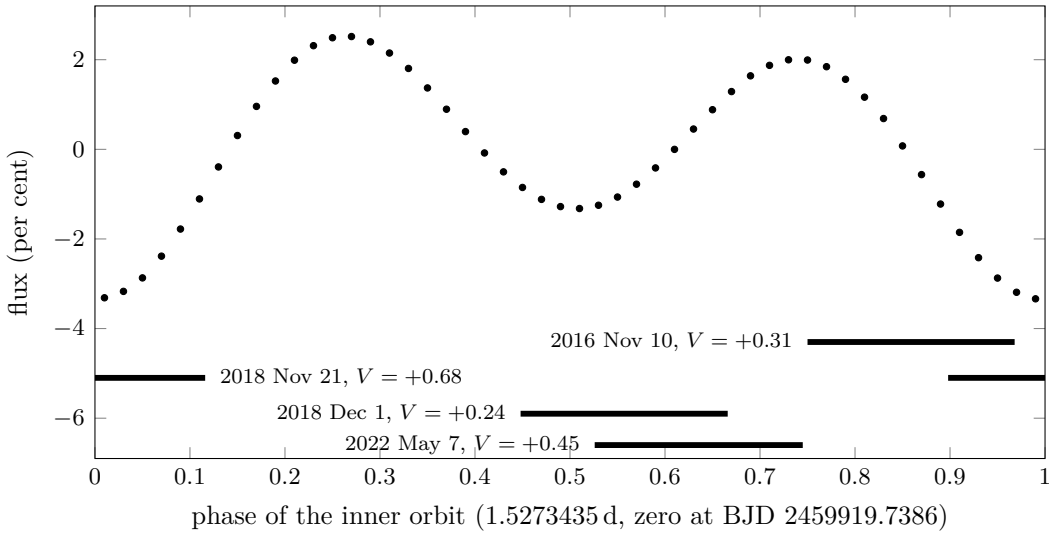



Figure 1: The TESS light curve of b Per in sector 19 (2019), folded on the ephemeris of the inner orbit and binned in 50 phase bins (errors smaller than the points); the bars mark the phases each 8 h LoTSS-DR3 observation covered, with Stokes V at the star's position in mJy beam$^{-1}$.

$2.3 \times 10^{15}$ erg s$^{-1}$ Hz$^{-1}$ at gigahertz frequencies, 0.14 mJy at the star's distance, below the survey's reach.

*4.3. The inner orbit*

The TESS fit gives a period of $1.5273435 \pm 0.0000002$ d, with phase zero, the deeper minimum, at BJD 2459919.73857. In sector 19 (Figure 1) the curve is a double wave 5.9 per cent from peak to trough, with minima 3.34 and 1.32 per cent below the mean at phases 0.00 and 0.51 and maxima 2.52 and 2.00 per cent above it at 0.27 and 0.74. The harmonic at half the period holds at 2.31–2.34 per cent in all three sectors and the term at the period, which makes the minima unequal, at 0.95–1.14 per cent. The four LoTSS-DR3 epochs fall at phases 0.75–0.97, 0.90–0.12, 0.45–0.67 and 0.53–0.75, and Stokes V is positive in each, the largest value at the conjunction of phase zero.

## 5. Discussion

A circularly polarised fraction near one and a brightness temperature above $10^{11}$ K for any region smaller than the inner orbit point to coherent emission, and the electron-cyclotron maser is the coherent process a magnetised star sustains: at the fundamental of the gyrofrequency 144 MHz needs about 51 G where it forms, at the second harmonic about 26 G. The gigahertz emission of b Per, non-thermal and flaring [1, 2], and the X-ray source mark a magnetically active component.

The X-ray luminosity places that component. A main-sequence star of 0.35 to 0.6 $M_\odot$ emits at most about $10^{-3}$ of its bolometric luminosity in X-rays, 0.5 to $3 \times 10^{29}$ erg s$^{-1}$, and the system's $7.2 \times 10^{30}$ erg s$^{-1}$ is 20 to 140 times that; A stars, without coronae or strong winds, emit far less [15]. X-ray luminosities of $10^{30}$ to $10^{31}$ erg s$^{-1}$ belong to the evolved, tidally locked active components of RS CVn and Algol binaries, and in an Algol the low-mass companion of an A or B star is a subgiant that has given up most of its mass. b Per's inner companion would then be such a star, locked to the 1.53 d orbit, and its magnetosphere the source of the 144 MHz emission, as LOFAR

hears in RS CVn binaries [4]. The light curve's term at the period, which keeps the minima unequal by about 1 per cent in every sector, fits a companion that contributes light and is lit by the A star. In the other reading the A star, catalogued as an Si star [9], carries a magnetosphere whose electron-cyclotron maser is beamed across Earth, as in CU Vir [18]; synchronised to the orbit, its pulses would recur at fixed orbital phases, while the four epochs show Stokes V of one sign over two thirds of the orbit.

Three measurements decide between the readings. The inner orbit spans 0.31 mas at the star's distance, so very-long-baseline astrometry over one orbit places the radio emission on one star or the other. A spectropolarimetric measurement of the A star's longitudinal field would show whether it is magnetic. Time-resolved 144 MHz data, which the DR3 visibilities hold, would show whether the emission comes in pulses at fixed phases or varies as an active star's does. The A star 50 Cas carries a circularly polarised 144 MHz source of the same kind with no companion above about $0.17\,M_\odot$ on its 1.6 d cycle [19]; b Per shows such emission in a system whose active, locked companion is known.

## 6. Related work

b Per's radio emission was found at 8.085 GHz [1], its flares and triple nature described by Hill et al. [2], and it was among the close binaries of a centimetre-wave survey [21] and the radio stars observed to link the optical and radio reference frames [20]. At 144 MHz, V-LoTSS catalogued the circularly polarised sources of LOFAR's second release [5], among them RS CVn binaries whose emission is coherent [4], and the release's two magnetic chemically peculiar stars are detections in total intensity [6]; neither covers b Per. No earlier detection of b Per below 1 GHz is found in the literature.

## 7. Conclusion

b Per, a radio-flaring triple at 117 pc, carries a 144 MHz source in LoTSS-DR3, circularly polarised at $+0.44 \pm 0.09$ mJy beam$^{-1}$ over four epochs of 2016–2022 and at a fraction near one in the brightest, at a luminosity of $1.2 \times 10^{16}$ erg s$^{-1}$ Hz$^{-1}$ and a brightness temperature that requires coherent emission. Its inner orbit of 1.5273435 d holds a companion of at least $0.35\,M_\odot$, and its X-ray luminosity is that of an evolved, locked active star rather than a dwarf, which makes that companion's electron-cyclotron maser the reading the data favour; the A star's own magnetosphere is the alternative, and astrometry of the radio source over one orbit would settle it.

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